Linux-1

In 1969 Unix was developed in a language called C. It was a very popular OS at that time.

The kernel is the core of an operating system. It is the code that interfaces the hardware with the software.

Other UNIX-like systems were designed but none caught on, until 1991 when Linus Torvalds started work on the Linux kernel.

All linux OS are derived from the base kernel developed by Linus. Each version of the OS is called a linux distribution.

Some examples of linux distributions are Debian, Red Hat, Ubuntu, Linux Mint, Gentoo, Arch Linux and openSUSE.

When getting a distribution you have three branches to choose from: 1. Stable 2. Testing and 3. Unstable.

You want the Stable branch. The others may have issues, bugs and are in the process of adding features that may not work properly.

The Shell

This is a program used to type in commands that are sent to the OS to carry out. In Linux the most popular shell is called **bash(Bourne Again Shell)**. Others are available such as ksh, zsh, tsch etc.

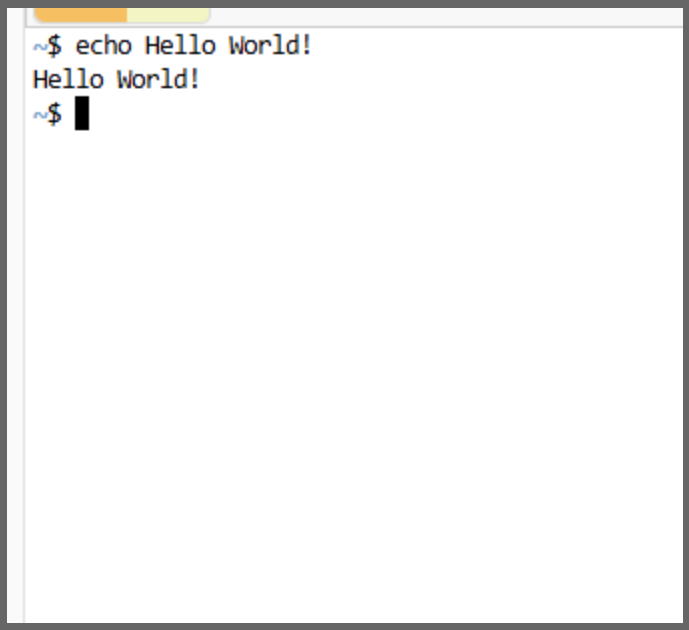
In the shell the **$** is used as a prompt. This means it's waiting for your command to be entered. Normally the default appearance of the shell prompt is:

```
username@hostname:current_directory $
```

Shell Commands

echo Hello World!

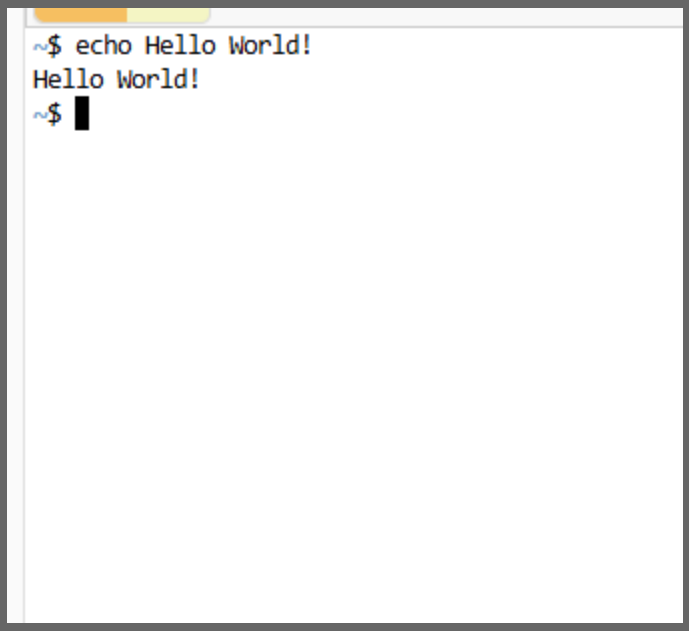
Enter the above command into the shell/terminal window and past a screenshot below of the result:



Explain what the echo command does:
Writes something in the command view thing

Try the following command (again take a screenshot of using the command and its result):

whoami



Explain what the whoami command does:

The “whoami” command displays the current user

Files and folders in all computers are stored in a hierarchical manner. This organization is normally displayed using an upside down tree where the top is called the **root**. In Linux the root is always /. The root is the container of all other files and folders in the computer. In Linux there

are common folders found directly in the root directory. Some of these include the **bin**, **etc** and **home** folders.

```
/
|-- bin
|   |-- file1
|   |-- file2
|-- etc
|   |-- file3
|   `-- directory1
|       |-- file4
|       `-- file5
|-- home
|-- var
```

Another name for a **folder** is a **directory**.

To locate anything in Linux you need the path to where it is in the Linux file system An example would be:

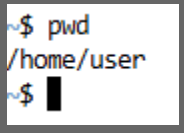
/etc/directory1/file4

This lets me know that **file4** is in a folder called **directory1** which in turn is in **etc** which in turn is in the **root**.

We often describe folders and files using a **parent child** relationship. In our last example **file4** is a child of **directory1** which in turn is a parent to **file4**. Folders that contain other files and folders are the parents to those items they contain.

To see the path to your current location use:
pwd

Include a screenshot of the above command and explain the path:

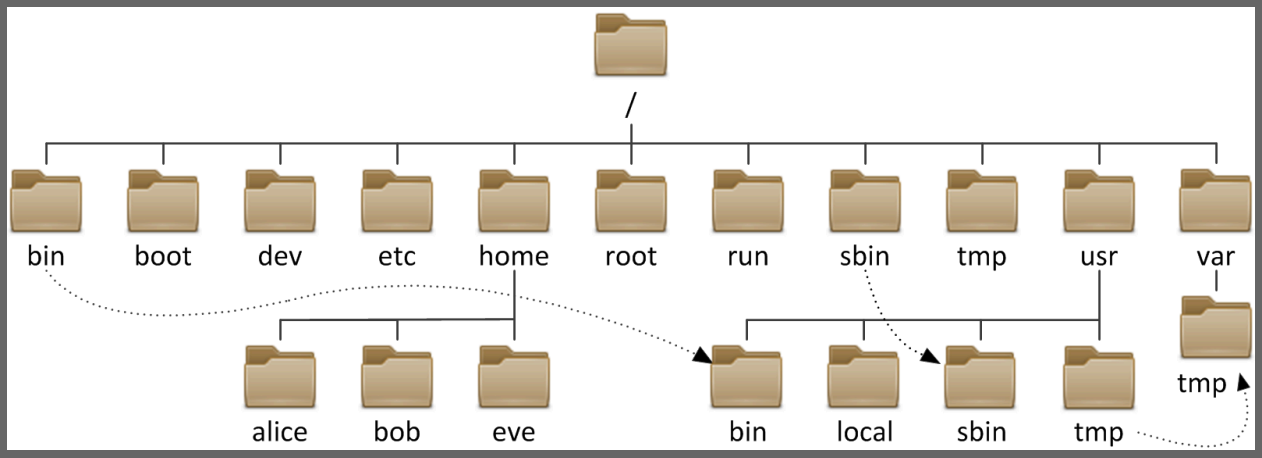


Im on cocalc so this is the default

/var

There are 2 types of paths:
Absolute: always starts from the root
Relative: relative to where you currently are and never starts with the root

Here is another example of a typical linux directory tree:



Question: What is the **absolute** path to the **home** directory?

Answer: /home

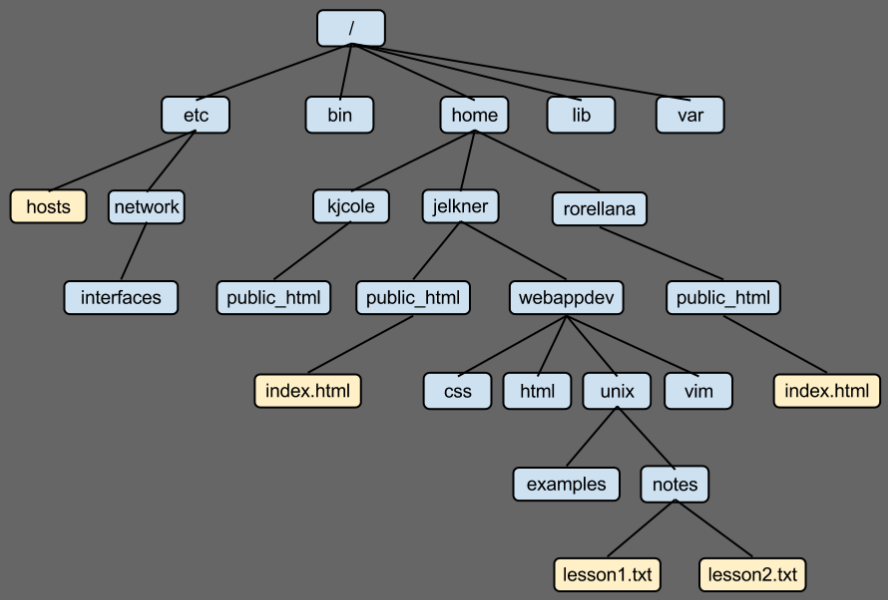
Question: What is the absolute path to sbin?

Answer: /usr/sbin or /sbin

Note that the file/directory is separated by a forward slash that looks like the root.

Practice:

Based on the following image of a linux computer directory tree find the absolute file paths to the listed locations:



Interfaces: /etc/network/interfaces

Index.html (the one on the far right side): /home/rorellana/public.html/index.html

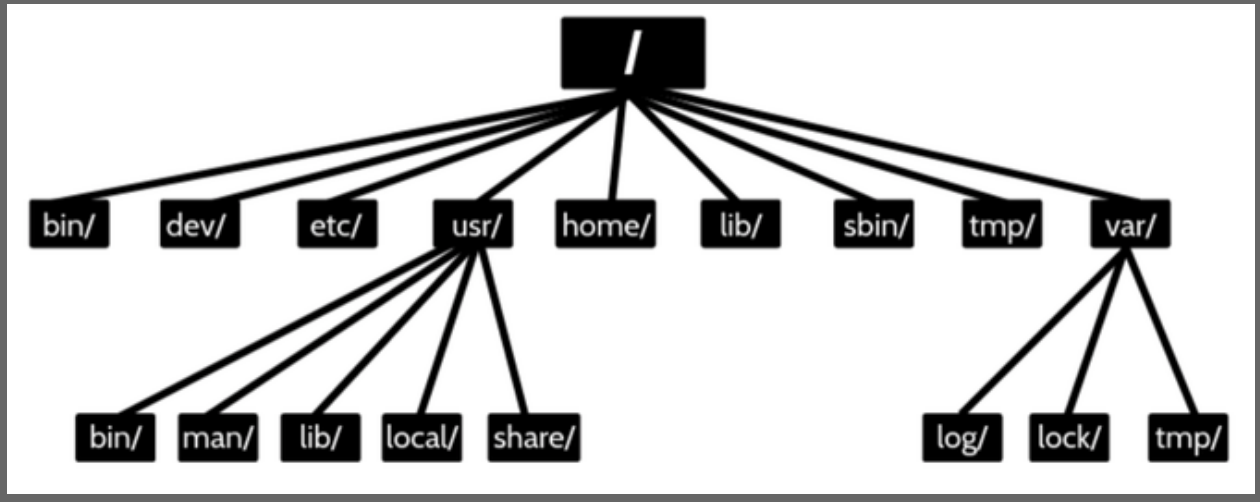
Lesson2.txt: /home/jelkner/webappdev/unix/notes/lesson2.txt

public_html found in **kjcole**: /home/kjcole/public.html

Hosts: /etc/hosts

The other file path system used is **relative**. This is more commonly used. Relative file paths are paths to files/directories relative to a starting point. In absolute paths the starting point is always the root. In relative file paths this can change.

Using the following linux directory tree we will write down the relative file paths to a file/directory from some other file/directory:



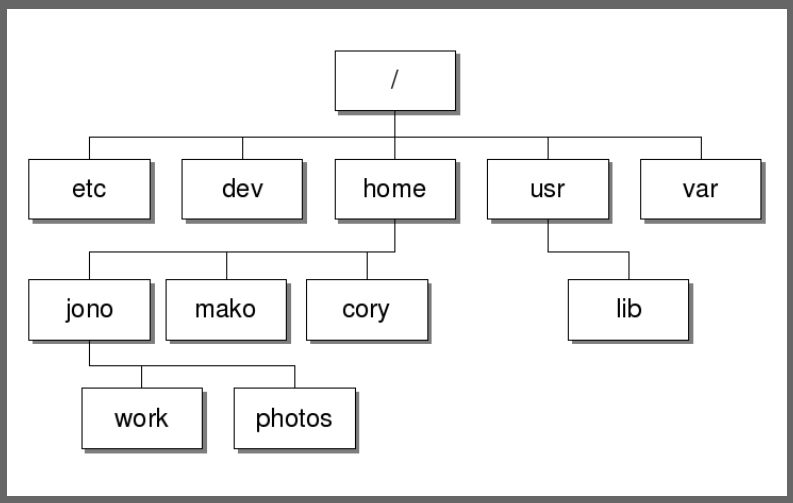
From the **root** to **bin**: bin or ./bin (note **./** means you are in the current directory)

From **root** to **lib is the /usr folder**: usr/lib or ./usr/lib

From **local** to to **etc**: ../../etc (note that we use **../** to move up out of a directory into a parent directory)

From **dev** to **tmp**: ../var/tmp or ../tmp

Practice: Use the following directory tree to practice relative paths:



From the **root** to **dev**: ./dev or dev

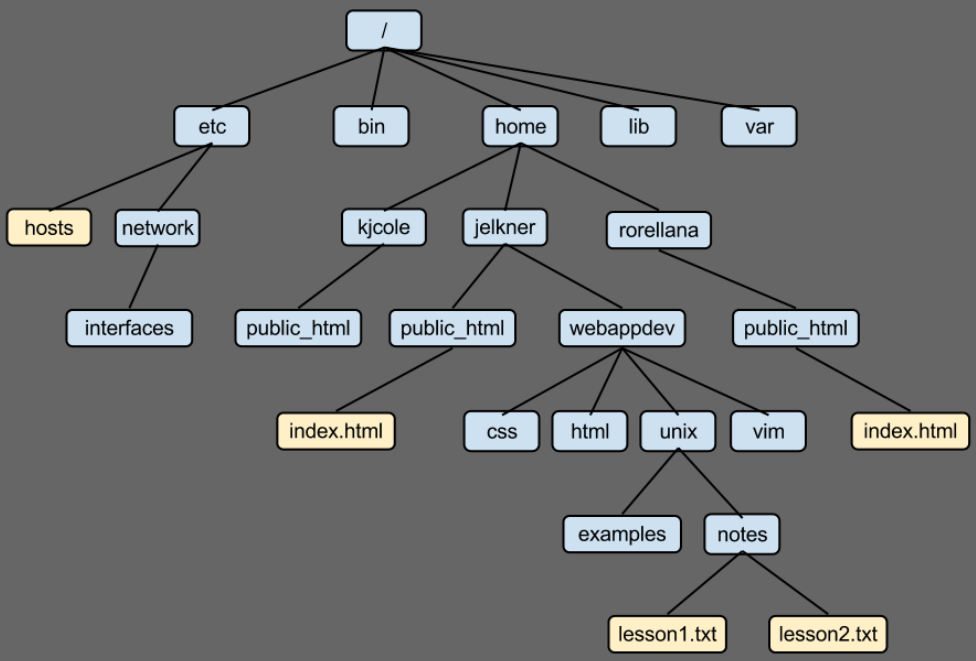
From **mako** to **work**: ../jono/work

From **lib** to the **root** directory: ../../

From **lib** to **usr**: ../

From **etc** to **usr**: ../usr

Some more practice using this tree:



From **network** to **hosts**: ../hosts

From **webappdev to lesson2.txt : /unix/notes/lesson2.txt**

From **css** to **vlm**: ../vim

The /home directory:

When you installed Linux you created a user account. Since you installed Linux you are what is called the superuser. You can then create other user accounts but none of them can do to your computer what you can, without permission. All users have a folder created for them in the /home directory. In the above tree you will notice that there are three users: **kjcole**, **rorellana** and **jelkner**. Each of these users have full permissions to do what they want in their own folders but nowhere else, unless of course, you are the superuser. To access your personal folder you can write the following command:

cd

Try using the above command. Next use the **pwd** command. Include a screenshot of the result below and explain what happens:

```
$ cd
$ pwd
/home/user
$
```

i am in my user folder

There are several other methods to get to your personal folder. They are as follows:

cd ~

cd /home/username

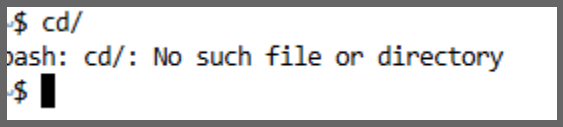
cd relative path to your folder

cd stands for change directory. You can include the **tilde ~** character which is shortcut for **/home/username** or you could just put in path to where you want to go after it i.e.

cd /etc.

Try the following and include a screenshot of the result (hit enter after each line):

cd /



The method of writing commands in a window to control a computer means you are using the CLI(___Command Line Interface__). Let's learn more of these commands.

The ls command lists all the contents of a directory. Type the following and include a screenshot:

ls



You could list the contents of any directory from where you are using paths. Try the following

ls /

Include a screenshot and explain what happens:



The ls / command in Linux lists the contents of the root directory

Most Linux commands contain optional **flags/switches/arguments**. These are additional pieces of info we can pass to a command to alter how it performs its actions. The ls command lists the contents of a directory but we can tell it to do more. For example there are many hidden files (normally start with a period i.e. .file1) in Linux (for security reasons) but we can show them as follows by using the **-a** flag:

ls -a

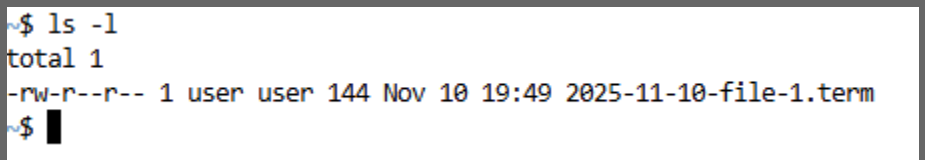
Try using this command in your personal directory. Include a screenshot below:

‘

Try the following command:

ls -l

This means to list all or everything The -l flag lists files and more details about them. You'll notice information on the read, write, execute privileges attached to each file, its datetime stamp, its size in bytes etc. Include a screenshot of the result below:

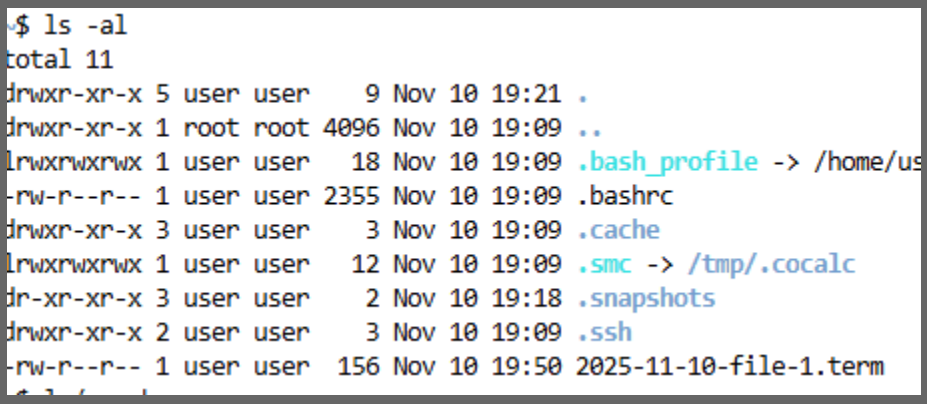


The information you normally would see is starting from the left: file permissions, number of links, owner name, owner group, file size, timestamp of last modification, and file/directory name.

You can add flags together to combine their functionality.

Try the following:

ls -al



The **touch** command creates empty files and can update its timestamp.

touch mysuperduperfile

You may not notice anything happening. Use the ls command to show the contents of where you are. What do you notice?

The file that was just created is now in the ls command

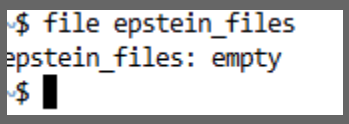
Since files in Linux don't have to have an extension it's difficult to see what it contains. In Windows data.txt contains text, ball.gif contains an image and so on. In Linux extensions are not necessary so data can hold an image or text and ball can hold text or an image. To find out what kind of file it is use the file command:

file filepath

Since you just made a file let's see what type it is by using the following command:

file mysuperduperfile

Include a screenshot of the result here:



Use the **ls -l** command to find the size of the file you just made. How big is it?

0 bytes

To add text to the file we have several options. A quick method is to use the following command:

echo data >> filepath

Try the following command:

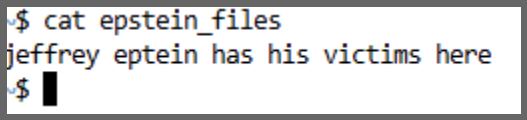
echo “This is a single line of text” >> mysuperduperfile

Now use **ls -l**. What is the new size of the file?

36 bits

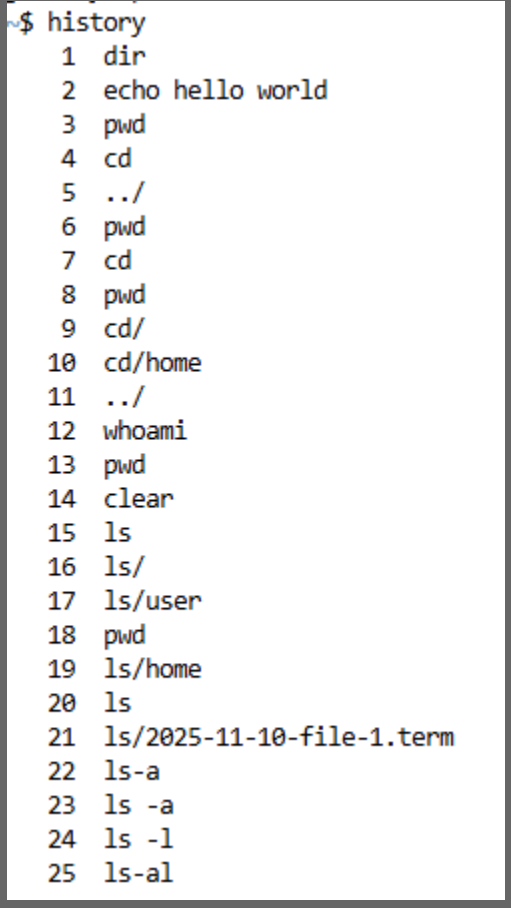
To quickly print the contents of a file to the terminal we can use the **cat** command. Try the following and include a screenshot:

cat mysuperduperfile



The history command shows you the history of all commands you entered. Try the following and include a screenshot:

history



```
~$ history
    1  dir
    2  echo hello world
    3  pwd
    4  cd
    5  ../
    6  pwd
    7  cd
    8  pwd
    9  cd/
   10  cd/home
   11  ../
   12  whoami
   13  pwd
   14  clear
   15  ls
   16  ls/
   17  ls/user
   18  pwd
   19  ls/home
   20  ls
   21  ls/2025-11-10-file-1.term
   22  ls-a
   23  ls -a
   24  ls -l
   25  ls-al
```

To scroll through previous commands hit the up arrow. Try pressing the UP ARROW.

The clear command clears the screen. Try typing in the following command:

clear

The mkdir command makes new empty folders. Make sure you are in your personal directory and type the following:

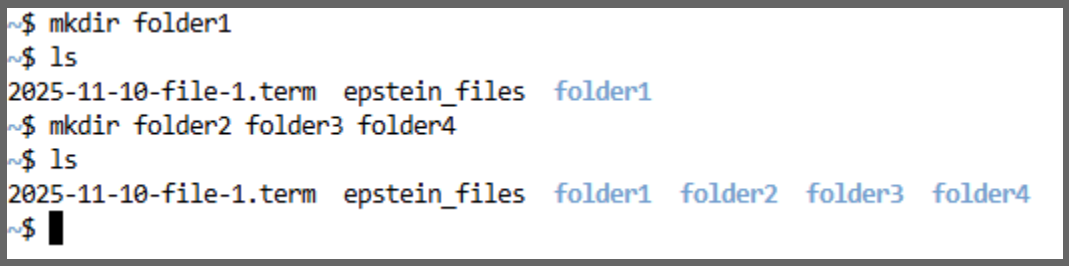
mkdir folder1

Nothing much happens but to see the changes type **ls**.

We can multiple folders at once by adding them to the command as follows (try it):

mkdir folder2 folder3 folder4

Type **ls** and include a screenshot:



To make subdirectories at the same time use the -p flag. Try the following:
mkdir -p folder5/folder6/folder7

Now type the following:

ls folder5

Now try

ls folder5/folder6

The **cp** command copies files. Try this:

cp mysuperduperfile folder1/mysuperduperfile

Now list the contents of the folder1 as follows:

ls folder1

What happened?

The file i copied was pasted into folder1

Moving a file is accomplished using the mv command. Try the following:

mv mysuperduperfile folder2

Lists the contents of your current directory and then list those of folder2. What happened?

The file got moved to folder 2

To remove files use the rm command. Let's create and then remove a file as follows:
touch file2.dat
ls
rm file2.dat
ls

To remove non empty directories use the -r flag:
Try the following:
rm folder5

What happens?

Now try:
rm -r folder5

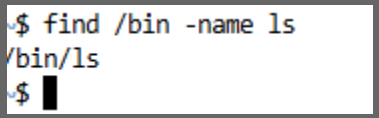
What happens?
It gets deleted

You could try rmdir as well for empty directories. Make an empty directory and then delete it. Include a screenshot:



Finding files is an important function so we have the find command for that. To find a file by name use the find command by telling it where to look as follows:
find /bin -name ls

Include a screenshot and explain what happens:



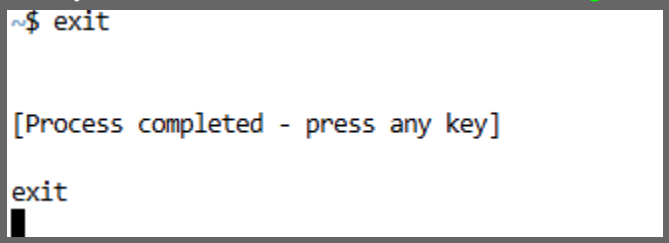
The command find /bin -name "ls" recursively searches the /bin directory and its subdirectories for a file named exactly "ls"

The **man** command lets you read the docs on a command too:
man ls

Gives a bunch of information about stuff

The **whatis** command gives a short description of a command:
whatis cat

Finally, to exit from the shell use the exit or logout command:



Questions and Exercises:

1. **Download and extract the following file using the commands (you'll need a wifi adapter→connect to YRDSB-Guest and then open browser and attempt to go to youtube.com→you should be redirected to a YRDSB login page where you will need to enter your credentials to access the internet)):**

   sudo apt install unzip
   wget https://github.com/aristocratos/btop/archive/refs/heads/main.zip
   unzip main.zip
   90
2. **The first line (sudo apt install unzip) installs the package unzip to help us with zipped files. The second line (wget https://github.com/aristocratos/btop/archive/refs/heads/main.zip) downloads a zip file off the internet. The third line (unzip main.zip) extracts the contents of the zip file into a folder named btop-main.**

3. **Access the btop-main folder and list all the 3 letter folders in btop-main.**
4. **Make sure you are in the folder btop-main. Use the pwd command and write down the current working directory:**

5. **Using the cd command and an absolute path get into the parent directory to btop-main.**

6. **Move back into btop-main and then use cd .. to get back into the parent directory.**

7. **Enter cd ~. Where are you now?**

8. **Use ls -l to find the largest file/folder in btop-main.**

9. Create a new file in btop-main called data.dat. Show its timestamp by using ls -l again.

10. Use the file command to find out what type of file README.md is. Write your findings here.

11. Use the cat command to display the contents of the README.md file.

12. This is a huge file to read so try using the less command to see less of it (press q to escape).

13. Try using the history command to scroll through the list of commands you've already used.

14. Clear your screen (use the clear command).

15. Copy the file README.md into your home directory using the cp command: cp README.md ~/README.md. Enter your home directory and find the file.

16. Copy the entire contents of the folder Img into a new folder called Images

17. Make a new directory in btop-main called myStuff. Move the README.md file into it.

18. Using the find command find the file logo.png in the Img folder (make sure you are in the btop-main folder.

19. Using the rm command delete the Images folder.

20. SSH stands for secure shell. It's a protocol that allows for secure remote access to a computer running a SSH server. Your teacher should have a secure shell server running on their computer. Each of you has been assigned a folder in your teacher's \home folder. The format of the folder is your firstNameLastName. To connect to a SSH server you will need three things: 1. Server IP/host address 2. Username 3. Password. Your username is your full name all in lowercase with no space in between. If you have an apostrophe or any other non alpha character it will be omitted. For example if your name is Josh Smith then your username is joshsmith. Your password is your student number. Ask your teacher for their SSH server's IP address (ex. 192.168.8.133).

    Make sure you are connected to the network running the ssh server (disconnect any wifi adapters connecting you to YRDSB-GUEST and reconnect via the wired network connection or using the wireless adapter connector to the network running the ssh server wirelessly…ask your teacher for its SSID and password → normally compTech___ and password compTech).

    Once you have all of this information you can gain access to your folder on the SSH server by typing the following into your terminal:
    ssh joshsmith@192.168.8.133

    Once connected you will be in /home/username. Type pwd to confirm this.

    Find the file password.txt and provide your teacher with the password.

You can exit an ssh connection by using the command exit.